

AMENDMENT NO. 1 JANUARY 1989

TO

IS : 12015 - 1987 SPECIFICATION FOR CYPERMETHRIN, TECHNICAL

(Page 4, Table 1) — Insert the following after Sl No. (iii) under the respective columns:

(1)	(2)	(3)	(4)	(5)
iv)	3-phenoxybenzaldehyde (MPB) content, percent by mass, <i>Max</i>	2.0	B	—
v)	<i>Cis-trans</i> ratio, percent of nominal value	<i>Cis Min</i> 40.0 <i>trans Max</i> 60.0	C	—

(Page 10) — Insert the following appendices after A-3.5:

A P P E N D I X B

[Table 1, Item (iv)]

DETERMINATION OF 3-PHIENOXYBENZALDEHYDE (MPB)

B-0. PRINCIPLE

B-0.1 A GLC unit with FI detector is used for this determination. Using solutions containing known amounts of the standard 3-phenoxybenzaldehyde (MPB), sample and the internal standard (*Is*), the response factor *R*, for MPB in the (*Is*) approach is arrived at. A solution containing known mass of the cypermethrin sample (under investigation) and *Is* is injected subsequently into the GLC unit. The percentage of MPB in the sample is then computed by the standard relationship.

B-1. APPARATUS

B-1.1 Gas Liquid Chromatograph — Suitable for analysis when operated under the following suggestive operating conditions. These conditions can be varied provided standardization is done.

Column	6' × 1/4" glass column filled with 3 per cent. OV-225 coated on Chromosorb WHP (80-100 mesh)
Column temperature	210°C
Injection port	260°C
Detector	260°C
Carrier gas	Nitrogen at 40 ml/min

B-1.2 Volumetric Flask

B-1.3 Microsyringe

B-2. REAGENTS

B-2.1 Metaphenoxy Benzaldehyde — AR Grade.

B-2.2 Chloroform — AR Grade.

B-2.3 Di-*n*-Butyl Phthalate — Internal standard AR Grade.

B-3. PREPARATION OF STANDARDS AND CALIBRATION

B-3.1 Weigh out accurately 2.5 g of di-*n*-butyl phthalate into a 50-ml volumetric flask and make up to volume with chloroform. This will give a solution containing 50 mg/ml of the internal standard.

B-3.2 Weigh out accurately 2.5 g of standard metaphenoxy benzaldehyde into a 50-ml volumetric flask and make up to volume with chloroform. This will give a solution containing 50 mg/ml of metaphenoxy benzaldehyde. Pipette out 1.0, 2.0, 3.0, 4.0 and 5.0 ml of this metaphenoxy benzaldehyde solution into a clean dry 10-ml flask. Add 5 ml of *Is* solution in each flask. Make up the volume with chloroform in the first four cases.

B-3.3 Inject 0.2 μ l of the standard solutions described under B-3.2 into the GIC unit. From the integrator print out and note down the peak areas of the metaphenoxy benzaldehyde and internal standard peaks.

B-3.4 Procedure

B-3.4.1 Weigh out accurately 5.0 g of the cypermethrin sample into a 10-ml volumetric flask. Measure out 5 ml of the *Is* solutions and make up to the mark with chloroform.

B-3.4.2 Introduce 0.2 μ l of this sample solution into GIC unit. From the integrator print out, note down the peak areas of metaphenoxy benzaldehyde and internal standard and internal standard peaks. Compute the percentage of the metaphenoxy benzaldehyde content in the sample.

B-3.5 Calculation

$$\text{3-phenoxy benzaldehyde content,} = \frac{R \times A_1 \times m_1}{A_2 \times m_2} \times 100$$

percent by mass

where

R = response factor ;

A_1 = area of the 3-phenoxy benzaldehyde peak obtained while analyzing the sample;

m_1 = mass of the internal standard added;

A_2 = area of the internal standard peak; and

m_2 = mass of the sample taken for analysis.

APPENDIX C

[Table 1, Item (v)]

DETERMINATION OF THE CIS-TRANS ISOMER RATIO IN CYPERMETHRIN, TECHNICAL

C-1. HPLC METHOD

C-1.1 Principle — The sample is subjected to HPLC analysis and the isomer distribution arrived at from the area of the peaks representing each isomer.

C-1.2 Apparatus

C-1.2.1 High performance liquid chromatograph — equipped with a printer plotter cum integrator and U.V. detector. The suggestive HPLC operating conditions are given below and they can be varied, depending upon instrument, provided standardization is done :

Column	Silica 5 μ m 25 cm $\times$ 4.6 mm, stainless steel
Solvent system	1 percent (v/v) di-isopropyl ether in <i>n</i> -pentane
Detector	U.V. at 280 nm
Flow rate	0.6 ml/min
Pressure	980 psi
Sample size	10 μ l

C-1.2.2 Standard Volumetric Flasks

C-1.3 Reagents

C-1.3.1 Di-isopropyl Ether — AR Grade.

C-1.3.2 *n*-pentane — AR Grade.

C-1.4 Procedure

C-1.4.1 Weigh out accurately 1.0 g of the cypermethrin, technical sample into a 100-ml volumetric flask and make up to volume using *n*-pentane containing 1 percent (v/v) di-isopropyl ether. Inject 10 μ l of the solution into the HPCL equipment, and record the chromatogram. This will consist of four major peaks. This first two peaks represent the *cis*-isomer, while the rest two represent the *trans*-isomer. Calculate the total area of each of the two pairs of peaks from the data printed out by the printer plotter. Compute the *cis-trans* isomer ratio as indicated in C-1.5.

C-1.5 Calculation

$$\begin{aligned} \text{Percentage of } cis\text{-isomer in the cypermethrin, technical sample} &= \frac{\text{Total area of the first two major peaks} \times 100}{\text{Total area of all the four major peaks}} \\ \text{Percentage of } trans\text{-isomer in the cypermethrin, technical sample} &= \frac{\text{Total area of the third and fourth major peaks} \times 100}{\text{Total area of all the four major peaks}} \end{aligned}$$

(AI-CDC 6)